[METRIC]

MIL-A-10841D
9 November 1981
SUPERSEDING
MIL-A-10841C
1 October 1969

MILITARY SPECIFICATION

ANTIMONY, TECHNICAL (METRIC)

This specification is approved for use by all
Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one technical grade of antimony crystals or powder.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-366 - Sieve, Test

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

FSC 6810

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

(The Code of Federal Regulations is available from the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

UNIFORM FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, NW, Washington, DC 20036.)

ASTM STANDARDS

D1193 - Reagent Water

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Purity. Antimony shall contain no less than 98.0 percent by weight of antimony (Sb) when tested as specified in 4.2.4.1.

3.2 Moisture. Antimony shall contain no more than 0.1 percent by weight of moisture when tested as specified in 4.2.4.2.

3.3 Particle size characteristics. Antimony shall conform to the particle size characteristics of table I when tested as specified in 4.2.4.3.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract,

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TABLE I. Particle size characteristics

Sieve size	Minimum percent by weight passing
No. 80	99.5
No. 140	90.0
No. 270	75.0

the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the antimony produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

4.2.2 Sampling.

4.2.2.1 For examination of packaging. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For test. Sampling shall be conducted in accordance with table II. A representative specimen of approximately 250 grams (g) shall be removed from each sample container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken.

TABLE II. Sampling for test

Number of containers in batch or lot	Number of sample containers
2 to 25	2
26 to 150	3
151 to 1,200	5
1,201 to 7,000	8
7,001 to 20,000	10
Over 20,000	20

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4.2.3 Inspection procedure.

4.2.3.1 For examination of packaging. The sample unit shall be one filled shipping container ready for shipment. Sample shipping containers shall be examined for the following defects using an AQL of 4.0 percent defective:

(a) Contents per container not as specified
(b) Container not in accordance with applicable regulations
(c) Container damaged or leaking
(d) Marking incorrect, missing or illegible

4.2.3.2 For test. Approximately equal portions of all of the specimens taken in 4.2.2.2 shall be thoroughly mixed to form a composited specimen of no less than 500 g. The composited specimen shall be tested as specified in 4.2.4. Each test shall be conducted in duplicate analysis. Failure of either analysis of any test shall be cause for rejection of the lot represented.

4.2.4 Tests. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.4.1 Purity. Weigh to the nearest 0.1 milligram (mg) approximately 0.2 g of the specimen into a 500-milliliter (ml) Erlenmeyer flask containing 5 g of sodium bisulfate and 20 ml of concentrated sulfuric acid. Heat until solution is complete. If any sulfur is visible in the flask, heat vigorously over an open flame to volatilize it. Cool, carefully add 10 ml of water, mix, and cool again. Add 5 ml of concentrated phosphoric acid and 1 g of sodium sulfite cautiously. Heat for 20 minutes to expel the excess sulfur dioxide and cool again. Add 100 ml of water and 15 ml of concentrated hydrochloric acid. Cool in an ice bath to 5° to 10°C and then titrate with 0.1N potassium permanganate solution until a pink color appears which persists for 5 seconds. Run a blank test. Calculate the percent antimony as follows:

$$\text{Percent antimony} = \frac{6.088A\ (B - C)}{W}$$

where: A = Normality of the potassium permanganate solution,
B = Milliliters of potassium permanganate solution used to titrate the specimen,
C = Milliliters of potassium permanganate solution used to titrate the blank, and
W = Weight of specimen in grams.

4.2.4.2 Moisture. Dry a glass-stoppered weighing bottle in a vacuum oven at 50° to 55°C and a vacuum of approximately 635 millimeters (mm) of mercury for 1 hour, cool to room temperature in a desiccator, and weigh to the nearest milligram. Transfer approximately 10 g of the specimen to the weighing bottle, stopper, and weigh to the nearest milligram. Heat unstoppered in the vacuum

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oven at 50° to 55°C and a vacuum of approximately 635 mm of mercury for 2 hours. Stopper the bottle immediately upon removal from the vacuum oven, cool to room temperature in a desiccator, and weigh to the nearest milligram. Calculate the percent moisture as follows:

$$\text{Percent moisture} = \frac{100\ (A - B)}{W}$$

where: A = Weight of stoppered bottle and specimen before heating in grams,
B = Weight of stoppered bottle and specimen after heating in grams, and
W = Weight of specimen in grams.

4.2.4.3 Particle size characteristics. Weigh to the nearest 0.1 g three clean, dry sieves of the sizes specified in table I and conforming to RR-S-366. Nest the sieves in order of increasing fineness, with the coarsest sieve on top and place on a bottom pan. Weigh to the nearest 0.5 g approximately 200 g of the specimen and place it on the top sieve. Cover the assembly and shake for 3 minutes by means of a mechanical shaker geared to produce 300 ± 15 gyrations and 150 ± 10 taps of the striker per minute. Weigh to the nearest 0.1 g the material retained on each sieve and calculate the percent specimen passing through each sieve as follows:

$$\text{Percent passing No. 80 sieve} = \frac{100\ (W - A)}{W}$$

$$\text{Percent passing No. 140 sieve} = \frac{100\ (W - A - B)}{W}$$

$$\text{Percent passing No. 270 sieve} = \frac{100\ (W - A - B - C)}{W}$$

where: A = Weight of specimen retained on No. 80 sieve in grams,
B = Weight of specimen retained on No. 140 sieve in grams,
C = Weight of specimen retained on No. 270 sieve in grams, and
W = Weight of specimen in grams.

5. PACKAGING

5.1 Level C. A quantity of antimony as specified (see 6.2) shall be unit packed and packed in accordance with the applicable requirements of the Code of Federal Regulations and the Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable. Containers shall be designed to assure safe delivery to destination without loss of or damage to contents.

5.2 Marking. Containers shall be marked in accordance with the Code of Federal Regulations and shall show the following information:

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(a) Item name
(b) Net weight of contents
(c) Lot number
(d) Name of manufacturer
(e) Date of manufacture of contents (month and year)
(f) Percent purity
(g) Precautionary information:

DANGER

DUST HARMFUL IF INHALED OR SWALLOWED
CAUSES IRRITATION OF SKIN, EYES, NOSE, AND THROAT
HEAT OR ACID MAY PRODUCE HIGHLY TOXIC FUMES
DUST MAY EXPLODE UPON CONTACT WITH FLAME

Do not get in eyes, on skin, or on clothing.
Do not breathe dust; use only with adequate ventilation.
Wash immediately upon skin contact.
Keep cool and dry; keep away from open flame.

6. NOTES

6.1 Intended use. Antimony is intended for use in starter mixtures.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification
(b) Unit quantity required (see 5.1)

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with ASTM E29.

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Preparing activity:

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-A-10841D Antimony, Technical (Metric)

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